

Assignment 9

1. Consider the standard panel data model:

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it} = \beta_0 + \beta_1 x_{it} + \alpha_i + v_{it}, \text{ where } i = 1, 2, \dots, n, \text{ and } t = 1, 2.$$

In addition, α_i is the fixed effect correlated with the endogenous variable x_{it}

(a) Write down the first-differenced equation.

(b) Derive the $\hat{\beta}_{1,FD}$. [Note that $E(\Delta v_{it})$ may not be zero. To estimate the first-differenced equation, we include an intercept α_0 in the first-differenced equation.]

(c) State the assumptions required for an unbiased and consistent $\hat{\beta}_{1,FD}$. Prove that under these assumptions $\hat{\beta}_{1,FD}$ is indeed unbiased and consistent.

(d) Is this $\hat{\beta}_{1,FD}$ the best linear unbiased estimator? Explain.

2. Use the data in RENTAL.dta for this exercise.

The data for the years 1980 and 1990 include rental prices and other variables for college towns. The idea is to see whether a stronger presence of students affects rental rates. The unobserved effects model is where *pop* is city population, *avginc* is average income, and *pctstu* is student population as a percentage of city population (during the school year).

(a) Estimate the equation by pooled OLS and report the results in standard form. What do you make of the estimate on the 1990 dummy variable? What do you get for $\hat{\beta}_{pctstu}$?

The regression model used here is as follows.

$$\log(\text{rent}) = \beta_0 + \gamma \text{dummy}_{1990} + \beta_1 \log(\text{pop}) + \beta_2 \log(\text{avginc}) + \beta_3 \text{pctstu} + u$$

(b) Now difference the equation and estimate by OLS. Compare your estimate of β_{pctstu} with that from part (a). Does the relative size of the student population appear to affect rental prices?

(c) Obtain the heteroskedasticity-robust standard errors for the first differenced equation in part (b). Does this change your conclusions?